

Khawaja Yunus Ali University

School of Science and Engineering

Department of Computer Science and Engineering

Mid-term Examination: Spring Semester 2021

Program: B.Sc. in CSE (Batch – 11th) 1st Year 2nd Semester

Course Title: Physics-II

Course Code: PHY 1201

Time: 10 Minutes

Total Marks: 10 X 1 = 10

Student ID:	Batch:	Session:
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Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

- What is the value of molar gas constant?
 - 1.67 JK^{-1}
 - 2.22 JK^{-1}
 - 7.07 JK^{-1}
 - 8.314 JK^{-1}
 - 13.6 JK^{-1}
- How many adiabatic steps in the Carnot's cycle?
 - 2
 - 3
 - 4
 - 5
 - 6
- Thermometric properties of thermocouple thermometer is
 - electromotive force.
 - brightness.
 - pressure.
 - volume.
 - resistance.
- The pressure of the gas generated by the collision of
 - the atoms with each others.
 - the molecules with each others.
 - the molecules with the wall of the container.
 - the external air with the wall of the container.
 - the molecules with the external air molecules.
- What is the unit of mean free path?
 - N
 - sec
 - Hz
 - J
 - m

6. The total thermal energy of system with 4 degree of freedom is
- $\frac{1}{2}RT$.
 - RT .
 - $2RT$.
 - $\frac{1}{4}RT$.
 - $\frac{1}{8}RT$.
7. For isochoric process – (Where, P -Pressure, Q -Heat, S -Entropy, T -temperature, W -external work done)
- $\Delta P = 0$
 - $\Delta Q = 0$
 - $\Delta S = 0$
 - $\Delta T = 0$
 - $\Delta W = 0$
8. A heat engine after doing work in each cycle rejects 70% of heat absorbed from the source. What will be the efficiency of the engine?
- 70%
 - 66%
 - 48%
 - 30%
 - 23%
9. Which one is the alpha particle?
- ${}_1^1H$
 - ${}_1^2H$
 - ${}_1^3H$
 - ${}_0^1n$
 - ${}_2^4He$
10. What is the radius of the 1st orbit of hydrogen atom?
- $0.53A^0$
 - $1.2A^0$
 - $2.4A^0$
 - $3.3A^0$
 - $5.3A^0$

Part 2: SAQ

Time: 1 Hour and 20 Minutes

Total Marks: 20

(Answer Question No. 1 and any three from the rest.)

[The figures in the right margin indicate full marks for the respective question]

1. See the stem and answer the questions that follows:

Categories	C_p (JK ⁻¹)	C_v (JK ⁻¹)
A	29.085	20.775
B	37.24	28.00

- a) Define C_p . 1
- b) Find the value of γ for both categories. 2
- c) Identify and give the at least one example of both categories of gases. 2
2. a) State the principle of the temperature measurement. 2
- b) Which metal is used in resistance thermometers? Why? 1+2=3
3. a) What is refrigeration? 1
- b) Draw the block diagram for refrigeration mechanism from your own concept. 2
- c) Establish an equation of efficiency of it. 2
4. a) State Mayer hypothesis. 1
- b) Establish the relation between Gibb's free energy and Helmholtz free energy. 2
- c) Find the change in entropy in order to transform 5kg water of 100°C to vapour of 100°C? 2
5. a) What is the angular momentum postulate of the Bohr atom model? 2
- b) Show that the velocity of an electron in the 1st orbit in hydrogen atom is $(\frac{1}{137})c$, where ' c ' is the velocity of light. 3